Agentic AI with LLM-RAG-Based Coordination for Self-Improving Swarm Robotics, SLAM, and IoT

Related Works

[1] Zhang et al. (2024) proposed Cooperative Embodied Language Agents (CoELA), a decentralized multi-agent framework that integrates large language models with perception, semantic, episodic and procedural memory, communication, planning, and execution modules. The cooperative planning problem is formulated as an extension of a decentralized partially observable Markov decision process incorporating communication actions. The framework was evaluated in the ThreeDWorld Multi-Agent Transport (TDW-MAT) and Communicative Watch-And-Help (C-WAH) environments against hierarchical planning and multi-agent learning baselines. In TDW-MAT, cooperation between two GPT-4-based CoELA agents achieved an aggregate transport rate of 0.71, compared with 0.61 for two rule-based hierarchical planners. Ablation experiments demonstrated the importance of memory and language-model reasoning, although communication between AI agents did not consistently yield significant efficiency improvements. The authors identified limited spatial reasoning, inadequate awareness of low-level execution and occasional reasoning errors as important limitations. Furthermore, the primary evaluation considered two-agent settings with noise-free communication, leaving scalability and robustness under unreliable communication insufficiently established. These findings motivate further investigation into SLAM-grounded planning, persistent experience retrieval and communication-aware coordination in multi-robot systems.

[2] Mandi et al. (2023) introduced RoCo, a zero-shot multi-robot collaboration framework that employs large language models for dialogue-based task coordination, feedback-guided sub-task generation, and three-dimensional waypoint planning. The framework combines agent-specific language-model reasoning with environmental validation and centralized RRT-based multi-arm motion planning. Its effectiveness was evaluated using RoCoBench, a six-task manipulation benchmark covering sequential and parallel coordination, heterogeneous robot capabilities, and varying workspace overlap. Experimental results demonstrated task-dependent performance, with the dialogue-based method achieving success rates of 0.93 and 0.65 on Sort Cubes and Move Rope, respectively, compared with 0.70 and 0.50 for a centralized oracle planner. However, the centralized approach performed better on the remaining four benchmark tasks. The authors also demonstrated that environmental feedback can improve proposed plans and that language-model-generated waypoints can accelerate certain motion-planning operations. Nevertheless, the framework relies on accurate environmental observations in simulation, employs open-loop trajectory execution, and incurs computational overhead from repeated LLM queries. Although RoCo supports in-context replanning, persistent experience-based learning, collaborative SLAM, and communication-constrained swarm coordination are not established by its evaluation. The study therefore provides a relevant foundation for investigating feedback-aware, retrieval-supported coordination in more complex multi-robot environments.

[3] Kannan et al. (2024) proposed SMART-LLM, a large language model-based framework for multi-robot task planning that integrates task decomposition, coalition formation, skill-based task allocation, and execution through structured Python-style few-shot prompts. The framework generates executable plans for heterogeneous robot

teams by considering individual robot capabilities, environmental objects, and task precedence constraints. The authors evaluated SMART-LLM on 36 benchmark tasks in AI2-THOR, categorized into elemental, simple, compound, and complex scenarios, and compared multiple LLM backbones against random and rule-based allocation methods. Using GPT-4, the framework achieved success rates of 1.00, 0.62, 0.69, and 0.71 across the four task categories, respectively. Ablation experiments showed that removing coalition formation reduced the aggregate success rate from 0.75 to 0.60, highlighting the importance of explicit team formation. Nevertheless, the approach assumes predefined robot skills, exhibits variability in complex-task allocation, and does not establish dynamic task reallocation during execution. Although the framework demonstrates the feasibility of language-model-guided task planning, its evaluation does not address collaborative SLAM, persistent retrieval-based learning, or communication-constrained large-scale swarm coordination. These limitations motivate investigation into adaptive task planning informed by spatial knowledge, execution feedback, and accumulated mission experience.

[4] Chen et al. (2024) investigated the scalability of large language model-based multi-robot task planning by comparing centralized (CMAS), decentralized (DMAS), and two hybrid multi-agent architectures (HMAS-1 and HMAS-2). The proposed HMAS-2 architecture combines centralized action generation with feedback from robot-specific LLM agents, enabling iterative revision of proposed task assignments. The authors also evaluated different context-history strategies to address increasing token consumption during multi-agent planning. Experiments were conducted in four warehouse-inspired environments, with robot teams ranging from 4 to 32 agents depending on the task. HMAS-2 achieved success rates of 82.5%, 57.5%, 62.5%, and 100% in BoxNet1, BoxNet2, Warehouse, and BoxLift, respectively, outperforming the other evaluated architectures. However, the centralized framework required fewer API calls and tokens, demonstrating a trade-off between planning reliability and computational overhead. The study further found that retaining state-action history was more effective than omitting history or including complete dialogue histories. Despite these contributions, the framework assumes full environmental knowledge, primarily evaluates scalability in abstract simulation environments, and does not investigate collaborative SLAM, persistent cross-mission learning, or unreliable network communication. Consequently, the findings provide a basis for investigating hierarchical, retrieval-supported multi-robot coordination under uncertain spatial information and communication constraints.

[5] Lajoie and Beltrame (2024) introduced Swarm-SLAM, an open-source decentralized collaborative simultaneous localization and mapping framework designed to support scalable, flexible and communication-efficient multi-robot perception. The framework integrates neighbor management, inter-robot loop closure detection and robust pose-graph optimization within a ROS 2 architecture that accommodates LiDAR, stereo and RGB-D sensing. Its principal methodological contribution is a budget-constrained spectral prioritization strategy that selects informative inter-robot loop closure candidates by maximizing pose-graph algebraic connectivity, reducing redundant geometric verification and communication overhead. Evaluation on seven sequences from five public datasets demonstrated improved localization convergence relative to greedy candidate prioritization, while comparisons with distributed optimization approaches showed favorable accuracy and communication requirements in the reported settings. The authors also validated the framework using three heterogeneous robots communicating over an ad-hoc network in an indoor parking lot. Nevertheless,

the physical evaluation was limited to a small robot team, and the architecture relies on dynamically elected robots for certain optimization operations. Furthermore, although the framework establishes a practical basis for decentralized spatial perception, it does not investigate language-model-based task coordination, retrieval-augmented decision-making or persistent cross-mission adaptation. These observations motivate examining how collaborative SLAM outputs can be integrated with higher-level agentic coordination and selective experience retrieval while preserving communication efficiency.

[6] Tian et al. (2022) proposed Kimera-Multi, a fully distributed collaborative SLAM framework for robust trajectory estimation and dense metric-semantic 3D reconstruction in multi-robot environments. The system integrates visual-inertial odometry, distributed place recognition, robust pose-graph optimization using distributed graduated nonconvexity (D-GNC), and local mesh deformation to construct geometrically consistent maps annotated with semantic information. The framework was evaluated using photorealistic simulations, EuRoC benchmarking datasets, and challenging outdoor robot datasets. Experimental results demonstrated that D-GNC achieved lower absolute trajectory errors than pairwise consistency maximization across the evaluated benchmark environments while maintaining accuracy comparable to centralized GNC optimization. The authors also reported communication reductions of up to 70% relative to a centralized keypoint-transmission baseline in the Vicon Room 2 dataset. Furthermore, semantic reconstruction accuracy for the merged Camp mesh increased from 79.4% to 95.2% following mesh optimization. Despite these contributions, the experimental evaluation primarily assumed continuous inter-robot communication, involved relatively small robot teams, and exhibited increased optimization time for larger pose graphs. Moreover, the framework does not investigate language-model-based task coordination, retrieval-augmented reasoning, or persistent experience-driven adaptation. Therefore, Kimera-Multi provides a foundation for integrating distributed semantic spatial perception with higher-level agentic planning, while the effectiveness of such integration remains to be experimentally established.

[7] Shinn et al. (2023) proposed Reflexion, a verbal reinforcement learning framework that enables large language model-based agents to improve their decision-making through iterative self-evaluation and episodic memory without modifying the underlying model parameters. The framework comprises an Actor that generates actions, an Evaluator that assesses task outcomes, and a Self-Reflection model that transforms performance feedback into natural-language lessons for subsequent trials. The approach was evaluated across sequential decision-making, multi-hop question answering, and code-generation benchmarks, including ALFWorld, HotPotQA, and HumanEval. Experimental results demonstrated that Reflexion successfully completed 130 of 134 ALFWorld tasks over repeated attempts and achieved 91.0% pass@1 accuracy on HumanEval Python, compared with 80.1% for the reported GPT-4 baseline. Ablation experiments further demonstrated the importance of combining meaningful evaluation feedback with explicit verbal reflection. However, the authors acknowledged that the approach can become trapped in suboptimal strategies, depends on the reliability of generated feedback, and maintains only a limited sliding window of past reflections. Furthermore, its evaluation did not address collaborative robotic systems, spatial mapping, or communication-constrained environments. These findings motivate investigating retrieval-augmented episodic memory and feedback-driven adaptation for multi-robot coordination, where task outcomes, spatial uncertainty, and communication

performance could provide measurable signals for improving subsequent mission planning.

[8] Zhu et al. (2024) introduced Retrieval-Augmented Embodied Agents (RAEA), a multimodal retrieval-augmented framework designed to improve robotic manipulation through the reuse of experiences stored in an external policy memory bank. The proposed architecture integrates a multimodal policy retriever, which identifies relevant experiences using instruction and observation embeddings, with a Transformer-based policy generator that incorporates retrieved information through cross-attention. The memory bank is constructed using multi-embodiment robotic data and contains instructions, observations, actions, and proprioceptive states, enabling the framework to exploit experiences across different robot platforms. The authors evaluated RAEA on Franka Kitchen, Meta-World, ManiSkill-2, and real-world manipulation tasks. On ManiSkill-2, RAEA achieved higher success rates than the evaluated visual-backbone baselines across six manipulation tasks. Real-world ablation experiments further demonstrated the importance of multimodal information and retrieved robot-state representations, with success rates decreasing from 54% to 36% when action and proprioceptive information were excluded. Retrieval from multiple robotic embodiments also improved success from 48% to 54% compared with a Franka-only memory bank. Although these findings establish the value of retrieval-augmented policy learning for embodied agents, the framework primarily addresses manipulation and does not evaluate collaborative SLAM, distributed multi-robot task coordination, or persistent memory updates from autonomous mission outcomes. Consequently, the study provides a methodological foundation for investigating multimodal experience retrieval within self-improving multi-robot systems, while the effectiveness of integrating retrieved experiences with spatial mapping, agentic planning, and communication-aware coordination remains to be established.

[9] Belcak et al. (2026) presented a position paper advocating the use of small language models (SLMs) as computationally efficient alternatives to general-purpose large language models (LLMs) for specialized operations within agentic AI systems. The authors argued that many agentic workloads consist of repetitive, narrowly defined tasks that can be handled by appropriately specialized SLMs, while complex reasoning and open-ended interactions may continue to require larger models. They proposed a heterogeneous architecture in which language models of different capabilities are invoked according to task requirements, together with a six-stage LLM-to-SLM conversion procedure involving secure interaction logging, data curation, task clustering, model selection, task-specific fine-tuning, and iterative refinement. Through qualitative case studies of MetaGPT, Open Operator, and Cradle, the authors estimated that approximately 60%, 40%, and 70% of their respective LLM invocations could potentially be replaced by specialized SLMs. However, these estimates were not validated through controlled end-to-end replacement experiments, and the proposed conversion procedure was not evaluated in collaborative robotic environments. Furthermore, the study did not investigate distributed SLAM, retrieval-augmented spatial reasoning, or adaptive model allocation under robot-level computational and communication constraints. Nevertheless, its emphasis on modular specialization and resource-efficient inference provides an architectural motivation for investigating heterogeneous language-model coordination in resource-constrained multi-robot systems, where specialist models could support routine local decisions while more capable models address complex planning requirements.

[10] Wang et al. (2023) introduced Voyager, an LLM-powered embodied lifelong learning agent designed to autonomously explore, acquire reusable skills, and improve task execution in the Minecraft environment. The framework integrates three principal components: an automatic curriculum that generates progressively challenging exploration objectives, an expanding skill library that stores executable programs for subsequent retrieval, and an iterative prompting mechanism that incorporates environmental feedback, execution errors, and self-verification to refine generated behaviors. Unlike approaches requiring repeated model fine-tuning, Voyager achieves experience-driven behavioral adaptation through in-context learning and the accumulation of reusable skills without updating the underlying GPT-4 parameters. Experimental results demonstrated that Voyager acquired 3.3 times more unique items, explored 2.3 times longer distances, and achieved selected technology-tree milestones up to 15.3 times faster than previously evaluated approaches. The authors further demonstrated that its acquired skill library could be transferred to a new Minecraft world to support unfamiliar tasks. Nevertheless, the evaluation focused on a single agent operating in a simulated environment and did not investigate distributed multi-robot coordination, collaborative simultaneous localization and mapping (SLAM), or resource-constrained IoT communication. Furthermore, its skill-retrieval mechanism does not establish the effectiveness of retrieval-augmented spatial reasoning in physical robotic systems. Therefore, Voyager provides a methodological foundation for experience-driven skill acquisition and retrieval-based behavioral adaptation, motivating further investigation into shared skill memories, feedback-driven planning, and persistent learning in collaborative multi-robot environments.

[11] Bousmalis et al. (2023) introduced RoboCat, a self-improving generalist robotic manipulation agent capable of learning from heterogeneous experiences collected across multiple tasks and robot embodiments. The proposed framework employs a visual goal-conditioned decision Transformer that processes image observations, proprioceptive states, and target goal images to generate robot actions. Its self-improvement procedure involves fine-tuning a pretrained generalist policy on a limited number of human demonstrations, deploying the adapted policy to autonomously collect additional task trajectories, and incorporating the resulting experiences into an expanded dataset for training a subsequent generalist model. The authors evaluated RoboCat across 240 training tasks and 13 additional fine-tuning tasks, including experiments on 36 physical robot arms representing three different embodiments. Experimental results demonstrated successful adaptation to previously unseen tasks using 100–1,000 demonstrations, with KUKA gear-lifting success increasing from 56% to 86% when the number of demonstrations increased from 500 to 1,000. A controlled self-improvement experiment further showed that incorporating robot-generated trajectories improved performance over training with demonstrations alone on all four evaluated tasks. Nevertheless, the demonstrated improvement procedure required initial human supervision and offline model retraining, and the authors acknowledged that the process was not fully autonomous. Furthermore, the study primarily addressed manipulation in controlled environments and did not investigate collaborative SLAM, distributed multi-robot coordination, retrieval-augmented mission planning, or communication-constrained IoT deployment. Consequently, RoboCat provides empirical support for experience-driven model improvement in embodied agents, motivating investigation into how validated multi-robot mission experiences could support persistent adaptation while maintaining spatial consistency, coordination reliability, and computational efficiency.

[12] Rana et al. (2023) introduced SayPlan, a spatially grounded large language model-based task planning framework designed to generate executable long-horizon plans in large-scale robotic environments. The proposed approach employs hierarchical 3D scene graphs to represent floors, rooms, objects, spatial relationships, and object affordances, enabling the language model to access structured environmental information during planning. To improve scalability, SayPlan incorporates an LLM-guided semantic search mechanism that selectively expands and contracts scene-graph nodes, allowing the planner to focus on task-relevant spatial information without processing the complete environmental representation. The framework further integrates classical path planning to handle navigation between locations and an iterative replanning procedure that uses scene-graph simulation feedback to identify and correct infeasible actions. The authors evaluated the approach in two large-scale environments spanning up to three floors and 36 rooms, with physical mobile-manipulator demonstrations supporting the feasibility of executing spatially grounded plans. These findings demonstrate the potential of combining structured spatial representations, language-based task reasoning, and algorithmic verification for complex robotic planning. Nevertheless, the framework relies on preconstructed scene graphs and primarily addresses planning for an individual robot rather than collaborative mapping and coordination among multiple robots. Furthermore, its iterative replanning mechanism does not establish persistent cross-mission learning, and its semantic search procedure retrieves environmental context rather than accumulated mission experiences. Consequently, SayPlan provides a foundation for investigating the integration of collaborative SLAM-derived spatial representations with retrieval-augmented planning and experience-driven adaptation in distributed multi-robot systems, where communication constraints, localization uncertainty, and repeated mission performance require systematic evaluation.

[13] Ma et al. (2024) proposed a distributed multi-robot exploration framework that integrates deep reinforcement learning, learned communication, and privileged critic training to address the trade-off between exploration efficiency and communication bandwidth. The framework represents each robot's partially observed environment as an informative graph and employs attention-based graph encoding to generate compact, fixed-size messages that summarize task-relevant spatial information. A cooperation decoder integrates locally available observations with messages received from other robots, enabling decentralized waypoint selection without requiring continuous exchange of complete occupancy maps. During training, the authors employ a privileged Soft Actor-Critic architecture in which the critic accesses ground-truth environmental information to guide policy learning, while the deployed policy operates using partial observations. The framework was trained on 4,000 simulated indoor maps and evaluated on 100 unseen environments using teams of four and eight robots. For four-robot exploration at 95% coverage, the proposed approach reduced average upload communication from 656.2 MB to 5.7 MB compared with the map-sharing baseline, while increasing the worst-robot travel distance only marginally from 200.8 m to 205.7 m. When evaluated with eight robots without retraining, the learned-communication policy reduced travel distance from 182.2 m to 168.9 m relative to the conventional baseline. Ablation experiments further indicated a 23.8% training-performance improvement from privileged critic information. Nevertheless, the evaluation was restricted to simulated environments, and the authors identified intermittent communication, further communication reduction, and physical robot validation as future directions. Furthermore, the framework does not investigate collaborative SLAM

pose optimization, language-model-based task planning, retrieval-augmented mission memory, or persistent adaptation across missions. These findings motivate exploring communication-aware agentic coordination that combines compact spatial representations with selective experience retrieval and collaborative mapping while systematically evaluating bandwidth consumption, exploration efficiency, and localization accuracy.

[14] Jadhav et al. (2024) proposed WiSER-X, a decentralized multi-robot exploration framework that exploits wireless signal-based relative-position sensing to coordinate autonomous robots without explicitly exchanging occupancy maps or task-allocation information. The framework integrates local LiDAR-based SLAM, frontier-based exploration, and range-and-bearing measurements derived from Ultra-Wideband and Wi-Fi signals. A probabilistic data association filter supports relative-position estimation under measurement uncertainty, while a hierarchical spatial data structure, termed HGrid, maintains historical estimates of neighboring robot positions to infer previously explored regions. These estimates are incorporated into frontier utility calculations to reduce redundant exploration and enable asynchronous mission termination without a centralized global map. The authors evaluated WiSER-X in a 1,600 m² simulated office environment using three robots over 20 trials per scenario and conducted hardware experiments with two customized TurtleBots in a 70 m² indoor environment. Simulation results demonstrated a 58% reduction in coverage overlap relative to independent exploration, while producing only 23% more overlap than a full-information-sharing baseline. The framework also achieved 1.65 times faster exploration completion than independent exploration and demonstrated adaptive behavior under reduced robot speed and complete robot failure. Nevertheless, the evaluation involved small robot teams, assumed known environment boundaries, and did not establish fully onboard execution of the complete framework during physical experiments. Furthermore, WiSER-X employs independent local SLAM rather than collaborative pose-graph optimization and does not investigate language-model-based task coordination, retrieval-augmented mission memory, or persistent cross-mission learning. Consequently, the study demonstrates the potential of implicit wireless sensing for communication-efficient decentralized exploration and motivates further investigation into combining relative sensing, collaborative SLAM, and selective experience-based coordination in adaptive multi-robot systems.

[15] Calzolari et al. (2024) proposed a decentralized multi-robot exploration framework that integrates multi-agent reinforcement learning, density-based frontier search, and explicit inter-robot communication to improve collaborative mapping in unknown environments. The exploration problem is formulated as a decentralized partially observable Markov decision process in which each robot independently selects among movement, stationary, and communication actions. To support scalable observation processing, the authors introduced a fixed-length frontier-point reachability representation derived from A*-estimated trajectories, capturing frontier density and navigation-distance statistics for candidate movement directions. The framework employs Heterogeneous-Agent Proximal Policy Optimization under centralized training and decentralized execution, with agent-specific policies and a shared training critic. Communication is constrained by robot proximity and requires explicit action selection, after which participating robots exchange and merge their reconstructed occupancy maps. The authors evaluated multilayer perceptron and convolutional neural network policies using two reward formulations across 200 randomized simulation

environments and validated the framework with four TurtleBot3 robots in a 6 m × 6 m laboratory arena. Simulation results showed that all evaluated policies exceeded 80% exploration coverage after 150 steps, while the best configuration reached up to 99% coverage. In physical experiments, the merged map approached complete coverage after approximately 60 steps. The reward-function analysis further demonstrated that excessive communication could reduce active exploration, emphasizing the importance of balancing information sharing and movement. Nevertheless, the experimental evaluation involved a small robot team and a limited physical environment, with localization supported by an external motion-capture system. Moreover, the framework does not investigate collaborative SLAM pose optimization, retrieval-augmented task reasoning, persistent cross-mission learning, or systematic network-bandwidth constraints. Consequently, this study provides a methodological foundation for communication-aware action selection in decentralized exploration and motivates further investigation into integrating selective map exchange with agentic planning, experience retrieval, and robust collaborative localization.

[16] Wang et al. (2023) proposed a bio-inspired decentralized multi-robot exploration framework that integrates multi-agent reinforcement learning with an implicit pheromone-based interaction mechanism to support cooperation in unknown environments under communication constraints. The proposed method, termed Multi-Agent Proximal Policy Optimization–Pheromone Interaction Mechanism (MAPPO-PIM), models pheromone release and its influence on robot movement to establish an indirect interaction mechanism inspired by collective behaviors observed in natural swarms. By incorporating pheromone feedback into reinforcement-learning reward shaping, the framework encourages robots to explore different unknown regions without requiring conventional explicit inter-robot communication. The authors compared MAPPO-PIM against baseline exploration methods and reported improvements of up to 45% in collaboration performance and a 19% reduction in execution time. These findings indicate that bio-inspired implicit coordination can improve cooperative exploration efficiency in the evaluated settings. Nevertheless, the available evidence does not establish the method's effectiveness for collaborative SLAM optimization, language-model-based task reasoning, retrieval-augmented experience reuse, or persistent cross-mission learning. Furthermore, the applicability of pheromone-based interaction to large-scale physical robot teams operating under realistic sensing and network constraints requires additional investigation. Consequently, this study provides a methodological basis for examining how lightweight bio-inspired coordination could complement higher-level agentic planning and spatially grounded decision-making in communication-constrained multi-robot systems.

[17] Tellaroli et al. (2024) proposed Frontier-Based Rendezvous (FBR), a decentralized exploration strategy designed to facilitate multi-robot rendezvous in initially unknown environments where communication is unavailable until robots come within mutual communication range. The method extends conventional frontier-based exploration by introducing an information-decay mechanism that generates virtual frontiers in previously explored regions, encouraging robots to revisit connecting areas and increasing the probability of unplanned encounters. Following a successful encounter, robots form connected clusters, exchange spatial information, elect a leader, and continue exploration using a merged cluster map and leader–follower coordination. The framework was evaluated using ROS/Gazebo simulations with TurtleBot3 robots

across five indoor environments and teams of three to eight robots. In three-robot experiments, the proposed approach reduced average rendezvous time from 1,591.70 s to 1,070.29 s in Campus-1 and from 1,370.68 s to 818.09 s in Campus-2 compared with standard frontier-based exploration. The authors also reported successful rendezvous across the evaluated FBR configurations, while the baseline failed to achieve rendezvous in some trials. The improvement was less pronounced with larger teams, and FBR was slightly slower than the baseline for eight robots in the Office environment. Nevertheless, the evaluation relied on simulated environments, homogeneous robots, and idealized formation control following rendezvous. Furthermore, the method employs independent local SLAM rather than distributed collaborative pose optimization and does not investigate language-model-based planning, retrieval-augmented mission memory, or persistent learning across missions. Consequently, FBR provides a useful foundation for rendezvous-aware exploration under severe connectivity constraints and motivates investigation into adaptive strategies that balance exploration progress, physical connectivity restoration, and selective exchange of spatial and experiential information.

[18] Wang et al. (2025; revised 2026) proposed COMRES-VLM, a vision-language-model-based framework for coordinated multi-robot exploration and semantic object search in unknown indoor environments. The framework adopts centralized high-level coordination with decentralized execution, aggregating individual robot occupancy maps into a global spatial representation. Frontier clusters and topological skeletons are extracted from the shared map and combined with robot states and optional natural-language priors to support joint waypoint assignment through a pretrained GPT-4o vision-language model. To improve execution robustness, the framework separates high-level reasoning from local A* navigation and employs waypoint validation, periodic replanning, and nearest-frontier fallback behavior. The authors evaluated the framework against RACER and AEP with Dynamic Voronoi Cells across small, medium, and large simulated indoor environments using teams of two, four, and six robots, respectively. In large environments, COMRES-VLM reduced average exploration completion time from 359.85 to 323.02 timesteps relative to AEP+DVC, corresponding to a 10.2% reduction. With informative natural-language target-location priors, search completion time decreased from 365.94 to 162.24 timesteps, a reduction of 55.7%. An ablation study further demonstrated that topological skeleton information supports exploration, whereas semantic priors substantially improve target-directed search; notably, additional frontier annotations could reduce search efficiency when informative priors were already available. Physical feasibility was demonstrated using a heterogeneous team comprising a quadruped and a quadrotor, although quantitative hardware comparisons were not reported. Despite these contributions, the framework relies on centralized foundation-model inference and shared map aggregation, introducing computational and communication dependencies. It does not establish bandwidth-constrained distributed SLAM optimization, retrieval-augmented experience reuse, or persistent learning across missions. Accordingly, COMRES-VLM provides an important foundation for integrating spatially grounded foundation-model reasoning with multi-robot coordination and motivates further investigation into hierarchical, communication-aware, and experience-driven exploration systems.

[19] Chiun et al. (2025) introduced MARVEL, a multi-agent reinforcement learning framework for collaborative exploration using robots equipped with constrained field-of-view sensors. Unlike conventional exploration planners that primarily optimize robot positions, MARVEL jointly selects navigation waypoints and sensor orientations to maximize informative environmental observations. The framework represents navigable space as a collision-free graph and combines frontier distributions, robot occupancy, spatial relationships, and heading-specific features using an attention-based encoder–decoder architecture. An information-driven action-pruning strategy reduces the number of candidate waypoint–orientation combinations, while a privileged attention-based critic supports cooperative policy learning through Soft Actor-Critic. The authors trained the framework using 5,663 randomly generated environments and evaluated it on 100 unseen maps measuring 90 m × 90 m, with robot teams of two, four, and eight agents. Under a 120° field of view and 10 m sensing range, MARVEL reduced the average trajectory length required for 99% coverage from 416.50 to 357.50 relative to the strongest baseline in the four-robot configuration, corresponding to an improvement of approximately 14.2%. The proposed method also achieved a 100% success rate across the evaluated team sizes and generalized to different sensor field-of-view and range configurations without retraining. Hardware feasibility was demonstrated using four Crazyflie drones, although sensing coverage was simulated from their real-time positions. Nevertheless, the framework assumes perfect inter-robot communication and primarily addresses two-dimensional exploration rather than fully distributed collaborative SLAM under network constraints. Furthermore, it does not incorporate language-model-based mission planning, retrieval-augmented experience reuse, or persistent learning across missions. Consequently, MARVEL provides a useful foundation for perception-aware multi-robot exploration and motivates further investigation into integrating sensor-aware planning with communication-efficient collaborative mapping and experience-driven agentic coordination.

[20] Chen et al. (2024) proposed the Transformer-Based Decentralized Exploration Network (TBDE-Net), a multi-agent deep reinforcement learning framework designed to improve exploration efficiency in unknown environments. The approach integrates Transformer-based map feature extraction with Multi-Agent Deep Deterministic Policy Gradient under centralized training and decentralized decision-making. Each robot processes a multi-channel occupancy-map representation containing spatial observations, candidate frontier locations, its current position, and available teammate information. A Transformer encoder extracts spatial relationships, while an LSTM-based policy component retains information obtained through previous communication events. The learned policy directly selects frontier locations as exploration targets, and a reward function encourages newly discovered area while penalizing repeated coverage, exploration time, and unnecessary decisions. The authors evaluated TBDE-Net against nearest-frontier, Next-Best-View, planning-based, and deep reinforcement-learning baselines in simulated environments using teams of two, three, and four robots. In the four-robot comparative experiment, TBDE-Net reduced average exploration time from 24.4 s to 21.3 s and total travel distance from 95.2 m to 91.9 m relative to the DME-DRL baseline. Additional ROS/V-REP simulations incorporating robot dynamics demonstrated improved exploration efficiency compared with nearest-frontier planning. Nevertheless, the authors acknowledged that the fixed network input dimensions limit adaptation to environments of different sizes and that changing the

robot-team size requires retraining. Furthermore, the framework uses centralized map integration and does not evaluate communication volume, distributed collaborative SLAM optimization, retrieval-augmented mission reasoning, or persistent learning across deployments. Consequently, TBDE-Net provides evidence that attention-based spatial feature extraction can support effective multi-robot exploration policies, while motivating further research into variable-scale, communication-aware, and experience-driven coordination architectures.